

Ian Zhang

Curriculum Vitae

Contact Information

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Education

PhD in Statistics, University of Toronto

Sep 2026 -

- Advisors: Joshua Speagle, Thibault Randrianarisoa

MSc in Statistics, University of Toronto

Sep 2025 - Jun 2026

- Advisor: Nancy Reid

Honours BSc in Statistics/Mathematics, University of Toronto

Sep 2021 - Jun 2025

- Advisor: Jun Young Park
- Graduated with high distinction

Publications and Preprints

[†]: Corresponding Author;

*: Equal contribution

Published

1. Michael A. Kouritzin, **Ian Zhang**[†], Jyoti Bhadana, Seoyeon Park. Markov processes for enhanced deepfake generation and detection[J]. *AIMS Mathematics*, 2026, 11(4): 11731-11759. doi: 10.3934/math.2026483

Under Review

1. Aditya Khan, **Ian Zhang**

2026

Parametric Hotspot Analysis with the Getis-Ord Statistic

In Progress

1. **Ian Zhang**^{*,†}, Hanlong Chen^{*}, ..., Howard Chertkow, ..., Malcolm Binns *2026*
Redundancy-Aware Model Assessment for Component and Cluster Selection in Clinical Subtype Discovery

2. Hanlong Chen*, **Ian Zhang***, ..., Malcolm Binns 2026
Continuous Cognitive Heterogeneity in Dementia: A Severity Factor and an Age-Related Amnesic Axis Without Reproducible Subtypes
3. **Ian Zhang**[†], Finn Tran, Meredith Franklin 2026
Spatial Dependence Regularization for Improved Generalization in Climate Downscaling Models
4. **Ian Zhang**[†], Thibault Randrianarisoa 2026
Likelihood Tempering to Mitigate Prior Dominance in Variational Posteriors for Bayesian Neural Networks

Research Experience

- Graduate Research Assistant**, Baycrest Institute/University of Toronto 2025-Present
Supervisor: Malcolm Binns
- Summer Research Trainee**, McGill University 2025
Supervisor: Jianguo (Jeff) Xia
- Research Assistant**, University of Alberta 2023-2024
Supervisor: Mike Kouritzin

Teaching

Teaching Assistant

- UofT Statistical Sciences Research Program (UTSSRP)** Summer 2026
STA437: Methods of Multivariate Data Winter 2026
MAT137: Calculus with Proofs Fall 2025 & Winter 2026
STA302: Methods of Data Analysis I Fall 2025

Presentations

Conference Talks

- 2026 Joint Statistical Meetings** 2026
Will be presenting *RAMA: Redundancy-Aware Model Assessment for PCA-Based Clinical Subtype Discovery*.
- 2026 SSC Annual Meeting in Hamilton** 2026
Presented *Likelihood Tempering to Mitigate Prior Dominance in Variational Posteriors for Bayesian Neural Networks* in the New Frontiers in Bayesian Analysis session.
- 2025 SSC Annual Meeting in Saskatoon** 2025
Presented *Markov Processes for Enhanced Deepfake Generation and Detection* in the Probability session.

Third Joint SIAM/CAIMS Annual Meetings

2025

Presented *Markov Processes for Enhanced Deepfake Generation and Detection* in the Machine Learning and Stochastic Processes session.

Posters

The Thirteenth Annual Canadian Statistics Student Conference

2025

Presented *Comparing Exact and Approximate Inference for Spatial Autocorrelation Tests: A Power Analysis*.

Honours and Awards

Anna & Alex Beverly Memorial Fellowship

2025

Awarded by University College (University of Toronto) for general proficiency during my undergraduate degree.

NSERC Undergraduate Student Research Award (x2)

2023, 2024

Awarded by the University of Alberta to research Markov process methods for deepfake generation and detection.

Jack Zwaigenbaum Scholarship

2022

Awarded by University College (University of Toronto) for academic excellence in my first year of undergraduate studies.

Dean's List Scholar (x4)

2022, 2023, 2024, 2025

Awarded by the Faculty of Arts and Science (University of Toronto) for maintaining academic excellence every year of my undergraduate degree.

Grants

CSSC Travel Grant

2025

Awarded to full-time Canadian students presenting a poster or giving a talk at the Canadian Statistics Student Conference.

Services

Conferences

Session chair, 2025 Third Joint SIAM/CAIMS Annual Meetings

Chaired the “Machine Learning and Stochastic Processes” session.

Other

Citizenship: Canadian

Languages Spoken: English (native), Mandarin (native), Japanese (intermediate)

Memberships: Statistical Society of Canada